

HW1

- **Submission date: 01/31/2025 (11:59 PM)**
- Revise the topics discussed in Class 1 *thoroughly* before attempting the assignment.

1. Machine Learning Paradigms. (Reference: pp 3 – 17 in the textbook)

Given below is a list of tasks. Classify them into **Supervised, Unsupervised, or Reinforcement Learning**. If no learning is taking place, your answer should be “**No learning.**”

- Predicting the price of a house based on features like square footage, number of rooms, and location.
- Clustering customer data to identify different customer segments based on purchase behavior.
- A robot learns to navigate a maze by receiving a reward for reaching the goal and a penalty for hitting walls.
- A program identifies even and odd numbers by checking the remainder when dividing by 2.
- Labeling emails as spam or not spam based on their content.
- Grouping news articles into categories of sports, technology, and politics without having prior access to labeled data.
- Training an autonomous car to drive by receiving positive rewards for staying on the road and negative rewards for accidents.
- Recommending movies to users based on their previous ratings and preferences.
- Identifying different plant species by grouping them based on their leaf patterns.
- Teaching a robot to play chess by rewarding it for winning a game and penalizing it for losing.
- A navigation app calculating the shortest path between two points using Dijkstra’s shortest path algorithm.
- Predicting the likelihood of consumer churning based on their transaction history and account features.
- Segmenting an image into different regions (e.g., sky, road, buildings, etc.) without using predefined labels.
- A computer converting temperature from Celsius to Fahrenheit using the formula $C \times \frac{9}{5} + 32$.
- A spam filter that blocks emails that contain keywords (e.g., “lottery” or “prize”) based on manually defined rules.
- A program learns to play a video game by maximizing score through trial and error.
- Using data from a retail store to cluster products into categories based on customer purchase patterns.
- Diagnosing diseases in patients based on symptoms and their corresponding diagnoses.
- Training a robotic arm to pick up objects by receiving rewards only for successfully grasping items.
- A robot moving in a straight line until it encounters an obstacle, at which point it turns left or right using preprogrammed logic.

2. Min and argmin. (Reference: class notes and [Wikilink](#))

- Given a list of real numbers, say, $l = [-3, 5, -10, 0, 15.5, -2]$, explain the difference between the terms: min, argmin, max, and argmax.
- Write a Python function `extreme_args()` that takes a list as input and returns its argmin and argmax. You may upload either .py or .ipynb files.
- Given the function $h(x) = x^2$, and the list $\mathcal{H} = [h(x_1), \dots, h(x_6)]$, where $x_i, 1 \leq i \leq 6$ are the elements of the list l in part a, what are $\underset{y \in \mathcal{H}}{\operatorname{argmin}} y$, and $\underset{x \in l}{\operatorname{argmax}} h(x)$?